Instagram Reel Virality Probability

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1 Problem Description

In the past decade social media has become one of the greatest drivers of culture. What goes viral on social media is highly impactful to society and what is popular. Instagram is one of the largest of the social media platforms, making it one of the most influential drivers of culture. In the past decade, vertical short-form video platforms, such as Instagram Reels, have become one of the most popular ways of consuming social media content. Being able to go viral on social media is by extension the ability to shape culture. This is why it is important to determine which characteristics of an Instagram Reel are most associated with virality, and how does observing these characteristics change the probability that a Reel is viral? This is important because it shapes generations' culture and shows what is valued in our society.

2 Data Sourcing and Processing

2.1 Data Source

We used What Makes an Instagram Reel Go Viral Dataset from Kaggle. This dataset includes 100,000 synthetic samples based on real data and 26 features including posting time, video length, virality score, and more.

2.2 Preprocessing

We found the data set was clean and not missing anything. Some of the features we found where not relevant in helping us answer our quiestion such as reel_id and creator_id.

We choose features based on what we intuitively thought would contribute to reel virality. We also found some features are redundent such as likes, comments, and shares all capture the same thing as the singular feature of virality score. This allowed us to narrow down the features we needed to answer our question.

The features we selected were: creator_followers, hook_strength_score, hashtags_count, trending_audio, retention_rate, impressions, and virality_score.

We then converted the selected features into discrete binary variables to simplify the Bayesian Network's state space. For each numeric feature, we assigned a value of 1 when its value was at or above the chosen threshold and 0 otherwise. We used the 99.5th percentile of virality_score to define what makes a video viral. For the other numeric features, we used their respective dataset medians (the 50th percentile).

Preprint.

Binary variable	Original feature	Threshold for a value of 1
bin_viral	virality_score	99.5th percentile
strong_hook	hook_strength_score	Median (≈ 0.499020)
many_hashtags	hashtags_count	Median (15)
many_impressions	impressions	Median (49,917,048.5)
many_followers	creator_followers	Median (2,500,456)
trending_audio	trending_audio	True
high_retention	retention_rate	Median (≈ 0.501637)

The trending_audio feature was already Boolean, so we cast it directly to an integer, mapping true to 1 and false to 0. These transformations provide a binary target and binary evidence variables for probabilistic inference. Because values equal to a threshold are included, ties can make the proportion of samples labeled 1 exceed the nominal percentile fraction.

3 Modeling and Inference

3.1 Model Structure and Assumptions

With said relevant features we will make a Bayesian Network using that subset of the 26 features.

[TODO: Define the variables, dependency structure, assumptions, and parameters. Explain how the network represents the problem and how its conditional probability tables are estimated.]

3.2 Inference Algorithm

We intend to use approximate inference to estimate the probability that a Reel is viral given observed characteristics.

[TODO: Specify the implemented inference algorithm, evidence and query variables, algorithm settings, and implementation details.]

4 Results and Discussion

4.1 Experimental Configurations

[TODO: Compare multiple configurations, such as inference settings, feature subsets, discretization thresholds, or network structures. Describe the evaluation procedure and explain observed performance differences.]

4.2 Quantitative Results

[TODO: Report measured results with suitable metrics, tables, or figures. Explain the metrics and compare performance across configurations.]

4.3 Qualitative Insights

[TODO: Explain what the model captures about the data, how observing features changes virality probabilities, and whether learned conditional probability tables align with intuition.]

4.4 Convergence and Scalability

[TODO: Discuss training or inference stability, convergence, runtime, and computational cost as the dataset or network grows.]

5 Conclusion

[TODO: Summarize key findings and model performance. Discuss limitations, including the use of synthetic data, and propose extensions or improvements.]

6 Reflections & Contributions

6.1 Advice for Future Students

[TODO: Share practical suggestions and lessons learned from the project.]

6.2 Individual Contributions and Reflections

Iban Rosales *[TODO: Write 1–2 sentences describing your contributions and a brief individual reflection on the project.]*

Ethan Fang *[TODO: Write 1–2 sentences describing your contributions and a brief individual reflection on the project.]*

Conner Satawidinata *[TODO: Write 1–2 sentences describing your contributions and a brief individual reflection on the project.]*

Benjamin Miller *Wrote a lot of the project writeup and assisted with overall project ideas.*

6.3 Use of Generative AI

Gen AI was used to assist in the generation of Python and Latex code in order to translate our ideas into code. We manaully reviewed all code and outputs to verify it matches our intent.

7 References

[TODO: Cite all external papers, datasets, and tools used in the project. Replace this placeholder with the actual reference entries.]